

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',     # BatchTopK
    'capa': 6,                   # Layer del modelo OthelloGPT
    'juegos': 12500,             # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../experiments'),
    'extras': {
        'expansion': 8,         # d_sae / d_in = 8192 / 512
        'k': 50,                # número de features activas L0 promedio    k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 12500
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex1 → {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
extras_id: ex1
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_topk\12500games\ex1\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio **k**, pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	MSE + $\cdot$ L1	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `BatchTopK` definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':                1e-3,
    'train_batch_size_tokens': 4096,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}_{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size    = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"])
val_loader   = DataLoader(val_dataset,   batch_size=cfg["train_batch_size_tokens"])

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape':     [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto Muestras      Shape
Train   590,000 (737500, 512)
Validación 147,500
Total    737,500

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAELens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - L0 es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAELens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'k_aux': 'Total params'
},
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		50
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f'  Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_12500g_ex1_best.pt

=====

Época 1/1000 | Train MSE: 0.372835 | Val MSE: 0.250671 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.250671)

Época 2/1000 | Train MSE: 0.231792 | Val MSE: 0.219217 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.219217)

Época 3/1000 | Train MSE: 0.208998 | Val MSE: 0.202060 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.202060)

Época 4/1000 | Train MSE: 0.194714 | Val MSE: 0.191799 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 4 | Val MSE: 0.191799)
Época 5/1000 | Train MSE: 0.185186 | Val MSE: 0.184754 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 5 | Val MSE: 0.184754)
Época 6/1000 | Train MSE: 0.179071 | Val MSE: 0.180570 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 6 | Val MSE: 0.180570)
Época 7/1000 | Train MSE: 0.174802 | Val MSE: 0.177696 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 7 | Val MSE: 0.177696)
Época 8/1000 | Train MSE: 0.171751 | Val MSE: 0.175401 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 8 | Val MSE: 0.175401)
Época 9/1000 | Train MSE: 0.169538 | Val MSE: 0.174039 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 9 | Val MSE: 0.174039)
Época 10/1000 | Train MSE: 0.167743 | Val MSE: 0.172888 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 10 | Val MSE: 0.172888)
Época 11/1000 | Train MSE: 0.166268 | Val MSE: 0.171827 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 11 | Val MSE: 0.171827)
Época 12/1000 | Train MSE: 0.165003 | Val MSE: 0.170914 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 12 | Val MSE: 0.170914)
Época 13/1000 | Train MSE: 0.164005 | Val MSE: 0.170188 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 13 | Val MSE: 0.170188)
Época 14/1000 | Train MSE: 0.163090 | Val MSE: 0.169501 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.169501)

Época 15/1000 | Train MSE: 0.162308 | Val MSE: 0.169106 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 15 | Val MSE: 0.169106)
 Época 16/1000 | Train MSE: 0.161703 | Val MSE: 0.168644 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.168644)
 Época 17/1000 | Train MSE: 0.161137 | Val MSE: 0.168266 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 17 | Val MSE: 0.168266)
 Época 18/1000 | Train MSE: 0.160655 | Val MSE: 0.167853 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.167853)
 Época 19/1000 | Train MSE: 0.160230 | Val MSE: 0.167662 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.167662)
 Época 20/1000 | Train MSE: 0.159855 | Val MSE: 0.167450 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 20 | Val MSE: 0.167450)
 Época 21/1000 | Train MSE: 0.159547 | Val MSE: 0.167195 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 21 | Val MSE: 0.167195)
 Época 22/1000 | Train MSE: 0.159273 | Val MSE: 0.167149 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.167149)
 Época 23/1000 | Train MSE: 0.158965 | Val MSE: 0.166988 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 23 | Val MSE: 0.166988)
 Época 24/1000 | Train MSE: 0.158717 | Val MSE: 0.166749 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.166749)
 Época 25/1000 | Train MSE: 0.158495 | Val MSE: 0.166724 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 25 | Val MSE: 0.166724)
 Época 26/1000 | Train MSE: 0.158292 | Val MSE: 0.166493 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 26 | Val MSE: 0.166493)
 Época 27/1000 | Train MSE: 0.158114 | Val MSE: 0.166416 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 27 | Val MSE: 0.166416)
 Época 28/1000 | Train MSE: 0.157966 | Val MSE: 0.166318 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 28 | Val MSE: 0.166318)
 Época 29/1000 | Train MSE: 0.157792 | Val MSE: 0.166329 | Train L0: 50.0 | Val L0: 50.0
 Época 30/1000 | Train MSE: 0.157630 | Val MSE: 0.166183 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 30 | Val MSE: 0.166183)
 Época 31/1000 | Train MSE: 0.157518 | Val MSE: 0.166234 | Train L0: 50.0 | Val L0: 50.0
 Época 32/1000 | Train MSE: 0.157420 | Val MSE: 0.166009 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 32 | Val MSE: 0.166009)
 Época 33/1000 | Train MSE: 0.157260 | Val MSE: 0.165963 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 33 | Val MSE: 0.165963)
 Época 34/1000 | Train MSE: 0.157120 | Val MSE: 0.165822 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 34 | Val MSE: 0.165822)
 Época 35/1000 | Train MSE: 0.156977 | Val MSE: 0.165788 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 35 | Val MSE: 0.165788)
 Época 36/1000 | Train MSE: 0.156904 | Val MSE: 0.165732 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 36 | Val MSE: 0.165732)
 Época 37/1000 | Train MSE: 0.156813 | Val MSE: 0.165744 | Train L0: 50.0 | Val L0: 50.0
 Época 38/1000 | Train MSE: 0.156724 | Val MSE: 0.165599 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 38 | Val MSE: 0.165599)
 Época 39/1000 | Train MSE: 0.156634 | Val MSE: 0.165653 | Train L0: 50.0 | Val L0: 50.0
 Época 40/1000 | Train MSE: 0.156549 | Val MSE: 0.165579 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 40 | Val MSE: 0.165579)
 Época 41/1000 | Train MSE: 0.156477 | Val MSE: 0.165480 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 41 | Val MSE: 0.165480)
 Época 42/1000 | Train MSE: 0.156397 | Val MSE: 0.165505 | Train L0: 50.0 | Val L0: 50.0
 Época 43/1000 | Train MSE: 0.156338 | Val MSE: 0.165422 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 43 | Val MSE: 0.165422)
 Época 44/1000 | Train MSE: 0.156261 | Val MSE: 0.165395 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 44 | Val MSE: 0.165395)
 Época 45/1000 | Train MSE: 0.156217 | Val MSE: 0.165344 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 45 | Val MSE: 0.165344)
 Época 46/1000 | Train MSE: 0.156140 | Val MSE: 0.165389 | Train L0: 50.0 | Val L0: 50.0
 Época 47/1000 | Train MSE: 0.156063 | Val MSE: 0.165270 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 47 | Val MSE: 0.165270)
 Época 48/1000 | Train MSE: 0.156020 | Val MSE: 0.165444 | Train L0: 50.0 | Val L0: 50.0
 Época 49/1000 | Train MSE: 0.155976 | Val MSE: 0.165269 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 49 | Val MSE: 0.165269)
 Época 50/1000 | Train MSE: 0.155888 | Val MSE: 0.165212 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 50 | Val MSE: 0.165212)
 Época 51/1000 | Train MSE: 0.155830 | Val MSE: 0.165323 | Train L0: 50.0 | Val L0: 50.0
 Época 52/1000 | Train MSE: 0.155790 | Val MSE: 0.165166 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 52 | Val MSE: 0.165166)
 Época 53/1000 | Train MSE: 0.155724 | Val MSE: 0.165074 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 53 | Val MSE: 0.165074)

Época 54/1000 | Train MSE: 0.155696 | Val MSE: 0.165106 | Train L0: 50.0 | Val L0: 50.0

Época 55/1000 | Train MSE: 0.155662 | Val MSE: 0.165038 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 55 | Val MSE: 0.165038)

Época 56/1000 | Train MSE: 0.155589 | Val MSE: 0.165044 | Train L0: 50.0 | Val L0: 50.0

Época 57/1000 | Train MSE: 0.155544 | Val MSE: 0.165085 | Train L0: 50.0 | Val L0: 50.0

Época 58/1000 | Train MSE: 0.155515 | Val MSE: 0.165025 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 58 | Val MSE: 0.165025)

Época 59/1000 | Train MSE: 0.155472 | Val MSE: 0.165092 | Train L0: 50.0 | Val L0: 50.0

Época 60/1000 | Train MSE: 0.155419 | Val MSE: 0.164941 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 60 | Val MSE: 0.164941)

Época 61/1000 | Train MSE: 0.155391 | Val MSE: 0.164898 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 61 | Val MSE: 0.164898)

Época 62/1000 | Train MSE: 0.155349 | Val MSE: 0.164976 | Train L0: 50.0 | Val L0: 50.0

Época 63/1000 | Train MSE: 0.155326 | Val MSE: 0.164900 | Train L0: 50.0 | Val L0: 50.0

Época 64/1000 | Train MSE: 0.155274 | Val MSE: 0.164893 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 64 | Val MSE: 0.164893)

Época 65/1000 | Train MSE: 0.155231 | Val MSE: 0.164885 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 65 | Val MSE: 0.164885)

Época 66/1000 | Train MSE: 0.155196 | Val MSE: 0.164893 | Train L0: 50.0 | Val L0: 50.0

Época 67/1000 | Train MSE: 0.155181 | Val MSE: 0.164810 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 67 | Val MSE: 0.164810)

Época 68/1000 | Train MSE: 0.155131 | Val MSE: 0.164825 | Train L0: 50.0 | Val L0: 50.0

Época 69/1000 | Train MSE: 0.155111 | Val MSE: 0.164793 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 69 | Val MSE: 0.164793)

Época 70/1000 | Train MSE: 0.155078 | Val MSE: 0.164757 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 70 | Val MSE: 0.164757)
 Época 71/1000 | Train MSE: 0.155054 | Val MSE: 0.164759 | Train L0: 50.0 | Val L0: 50.0
 Época 72/1000 | Train MSE: 0.155016 | Val MSE: 0.164768 | Train L0: 50.0 | Val L0: 50.0
 Época 73/1000 | Train MSE: 0.154994 | Val MSE: 0.164730 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 73 | Val MSE: 0.164730)
 Época 74/1000 | Train MSE: 0.154966 | Val MSE: 0.164712 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 74 | Val MSE: 0.164712)
 Época 75/1000 | Train MSE: 0.154960 | Val MSE: 0.164756 | Train L0: 50.0 | Val L0: 50.0
 Época 76/1000 | Train MSE: 0.154906 | Val MSE: 0.164653 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 76 | Val MSE: 0.164653)
 Época 77/1000 | Train MSE: 0.154896 | Val MSE: 0.164651 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 77 | Val MSE: 0.164651)
 Época 78/1000 | Train MSE: 0.154877 | Val MSE: 0.164698 | Train L0: 50.0 | Val L0: 50.0
 Época 79/1000 | Train MSE: 0.154857 | Val MSE: 0.164677 | Train L0: 50.0 | Val L0: 50.0
 Época 80/1000 | Train MSE: 0.154820 | Val MSE: 0.164626 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 80 | Val MSE: 0.164626)
 Época 81/1000 | Train MSE: 0.154788 | Val MSE: 0.164590 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 81 | Val MSE: 0.164590)
 Época 82/1000 | Train MSE: 0.154779 | Val MSE: 0.164658 | Train L0: 50.0 | Val L0: 50.0
 Época 83/1000 | Train MSE: 0.154764 | Val MSE: 0.164600 | Train L0: 50.0 | Val L0: 50.0
 Época 84/1000 | Train MSE: 0.154734 | Val MSE: 0.164636 | Train L0: 50.0 | Val L0: 50.0
 Época 85/1000 | Train MSE: 0.154714 | Val MSE: 0.164656 | Train L0: 50.0 | Val L0: 50.0
 Época 86/1000 | Train MSE: 0.154694 | Val MSE: 0.164541 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 86 | Val MSE: 0.164541)
 Época 87/1000 | Train MSE: 0.154661 | Val MSE: 0.164576 | Train L0: 50.0 | Val L0: 50.0
 Época 88/1000 | Train MSE: 0.154639 | Val MSE: 0.164537 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 88 | Val MSE: 0.164537)

Época 89/1000 | Train MSE: 0.154617 | Val MSE: 0.164523 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 89 | Val MSE: 0.164523)

Época 90/1000 | Train MSE: 0.154596 | Val MSE: 0.164685 | Train L0: 50.0 | Val L0: 50.0

Época 91/1000 | Train MSE: 0.154586 | Val MSE: 0.164520 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 91 | Val MSE: 0.164520)

Época 92/1000 | Train MSE: 0.154562 | Val MSE: 0.164469 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 92 | Val MSE: 0.164469)

Época 93/1000 | Train MSE: 0.154535 | Val MSE: 0.164497 | Train L0: 50.0 | Val L0: 50.0

Época 94/1000 | Train MSE: 0.154518 | Val MSE: 0.164469 | Train L0: 50.0 | Val L0: 50.0

Época 95/1000 | Train MSE: 0.154490 | Val MSE: 0.164550 | Train L0: 50.0 | Val L0: 50.0

Época 96/1000 | Train MSE: 0.154481 | Val MSE: 0.164419 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 96 | Val MSE: 0.164419)

Época 97/1000 | Train MSE: 0.154472 | Val MSE: 0.164483 | Train L0: 50.0 | Val L0: 50.0

Época 98/1000 | Train MSE: 0.154448 | Val MSE: 0.164422 | Train L0: 50.0 | Val L0: 50.0

Época 99/1000 | Train MSE: 0.154429 | Val MSE: 0.164439 | Train L0: 50.0 | Val L0: 50.0

Época 100/1000 | Train MSE: 0.154402 | Val MSE: 0.164414 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 100 | Val MSE: 0.164414)

Época 101/1000 | Train MSE: 0.154386 | Val MSE: 0.164423 | Train L0: 50.0 | Val L0: 50.0

Época 102/1000 | Train MSE: 0.154376 | Val MSE: 0.164459 | Train L0: 50.0 | Val L0: 50.0

Época 103/1000 | Train MSE: 0.154354 | Val MSE: 0.164370 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 103 | Val MSE: 0.164370)

Época 104/1000 | Train MSE: 0.154338 | Val MSE: 0.164391 | Train L0: 50.0 | Val L0: 50.0

Época 105/1000 | Train MSE: 0.154321 | Val MSE: 0.164383 | Train L0: 50.0 | Val L0: 50.0

Época 106/1000 | Train MSE: 0.154307 | Val MSE: 0.164382 | Train L0: 50.0 | Val L0: 50.0

Época 107/1000 | Train MSE: 0.154297 | Val MSE: 0.164409 | Train L0: 50.0 | Val L0: 50.0

Época 108/1000 | Train MSE: 0.154269 | Val MSE: 0.164432 | Train L0: 50.0 | Val L0: 50.0

Época 109/1000 | Train MSE: 0.154287 | Val MSE: 0.164355 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 109 | Val MSE: 0.164355)

Época 110/1000 | Train MSE: 0.154264 | Val MSE: 0.164383 | Train L0: 50.0 | Val L0: 50.0

Época 111/1000 | Train MSE: 0.154247 | Val MSE: 0.164325 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 111 | Val MSE: 0.164325)

Época 112/1000 | Train MSE: 0.154223 | Val MSE: 0.164429 | Train L0: 50.0 | Val L0: 50.0

Época 113/1000 | Train MSE: 0.154214 | Val MSE: 0.164327 | Train L0: 50.0 | Val L0: 50.0

Época 114/1000 | Train MSE: 0.154193 | Val MSE: 0.164313 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 114 | Val MSE: 0.164313)

Época 115/1000 | Train MSE: 0.154176 | Val MSE: 0.164352 | Train L0: 50.0 | Val L0: 50.0

Época 116/1000 | Train MSE: 0.154158 | Val MSE: 0.164332 | Train L0: 50.0 | Val L0: 50.0

Época 117/1000 | Train MSE: 0.154150 | Val MSE: 0.164340 | Train L0: 50.0 | Val L0: 50.0

Época 118/1000 | Train MSE: 0.154141 | Val MSE: 0.164304 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 118 | Val MSE: 0.164304)

Época 119/1000 | Train MSE: 0.154119 | Val MSE: 0.164329 | Train L0: 50.0 | Val L0: 50.0

Época 120/1000 | Train MSE: 0.154122 | Val MSE: 0.164246 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 120 | Val MSE: 0.164246)

Época 121/1000 | Train MSE: 0.154103 | Val MSE: 0.164273 | Train L0: 50.0 | Val L0: 50.0

Época 122/1000 | Train MSE: 0.154087 | Val MSE: 0.164299 | Train L0: 50.0 | Val L0: 50.0

Época 123/1000 | Train MSE: 0.154084 | Val MSE: 0.164339 | Train L0: 50.0 | Val L0: 50.0

Época 124/1000 | Train MSE: 0.154075 | Val MSE: 0.164318 | Train L0: 50.0 | Val L0: 50.0

Época 125/1000 | Train MSE: 0.154066 | Val MSE: 0.164242 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 125 | Val MSE: 0.164242)

Época 126/1000 | Train MSE: 0.154051 | Val MSE: 0.164296 | Train L0: 50.0 | Val L0: 50.0

Época 127/1000 | Train MSE: 0.154038 | Val MSE: 0.164289 | Train L0: 50.0 | Val L0: 50.0

Época 128/1000 | Train MSE: 0.154025 | Val MSE: 0.164287 | Train L0: 50.0 | Val L0: 50.0

Época 129/1000 | Train MSE: 0.154022 | Val MSE: 0.164301 | Train L0: 50.0 | Val L0: 50.0

Época 130/1000 | Train MSE: 0.154007 | Val MSE: 0.164378 | Train L0: 50.0 | Val L0: 50.0

Época 131/1000 | Train MSE: 0.154000 | Val MSE: 0.164368 | Train L0: 50.0 | Val L0: 50.0

Época 132/1000 | Train MSE: 0.153991 | Val MSE: 0.164233 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_16_12500g_ex1_best.pt

Mejor modelo guardado (época 132 | Val MSE: 0.164233)

Época 133/1000 | Train MSE: 0.153978 | Val MSE: 0.164246 | Train L0: 50.0 | Val L0: 50.0
 Época 134/1000 | Train MSE: 0.153967 | Val MSE: 0.164280 | Train L0: 50.0 | Val L0: 50.0
 Época 135/1000 | Train MSE: 0.153953 | Val MSE: 0.164243 | Train L0: 50.0 | Val L0: 50.0
 Época 136/1000 | Train MSE: 0.153945 | Val MSE: 0.164264 | Train L0: 50.0 | Val L0: 50.0
 Época 137/1000 | Train MSE: 0.153945 | Val MSE: 0.164226 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 137 | Val MSE: 0.164226)
 Época 138/1000 | Train MSE: 0.153926 | Val MSE: 0.164240 | Train L0: 50.0 | Val L0: 50.0
 Época 139/1000 | Train MSE: 0.153911 | Val MSE: 0.164291 | Train L0: 50.0 | Val L0: 50.0
 Época 140/1000 | Train MSE: 0.153907 | Val MSE: 0.164345 | Train L0: 50.0 | Val L0: 50.0
 Época 141/1000 | Train MSE: 0.153903 | Val MSE: 0.164219 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 141 | Val MSE: 0.164219)
 Época 142/1000 | Train MSE: 0.153891 | Val MSE: 0.164249 | Train L0: 50.0 | Val L0: 50.0
 Época 143/1000 | Train MSE: 0.153884 | Val MSE: 0.164207 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 143 | Val MSE: 0.164207)
 Época 144/1000 | Train MSE: 0.153870 | Val MSE: 0.164403 | Train L0: 50.0 | Val L0: 50.0
 Época 145/1000 | Train MSE: 0.153869 | Val MSE: 0.164206 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 145 | Val MSE: 0.164206)
 Época 146/1000 | Train MSE: 0.153855 | Val MSE: 0.164180 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 146 | Val MSE: 0.164180)
 Época 147/1000 | Train MSE: 0.153854 | Val MSE: 0.164196 | Train L0: 50.0 | Val L0: 50.0
 Época 148/1000 | Train MSE: 0.153841 | Val MSE: 0.164165 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 148 | Val MSE: 0.164165)
 Época 149/1000 | Train MSE: 0.153844 | Val MSE: 0.164156 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 149 | Val MSE: 0.164156)
 Época 150/1000 | Train MSE: 0.153826 | Val MSE: 0.164184 | Train L0: 50.0 | Val L0: 50.0
 Época 151/1000 | Train MSE: 0.153822 | Val MSE: 0.164158 | Train L0: 50.0 | Val L0: 50.0
 Época 152/1000 | Train MSE: 0.153822 | Val MSE: 0.164153 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 152 | Val MSE: 0.164153)
 Época 153/1000 | Train MSE: 0.153807 | Val MSE: 0.164145 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 153 | Val MSE: 0.164145)
 Época 154/1000 | Train MSE: 0.153802 | Val MSE: 0.164161 | Train L0: 50.0 | Val L0: 50.0
 Época 155/1000 | Train MSE: 0.153788 | Val MSE: 0.164107 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 155 | Val MSE: 0.164107)
 Época 156/1000 | Train MSE: 0.153783 | Val MSE: 0.164154 | Train L0: 50.0 | Val L0: 50.0
 Época 157/1000 | Train MSE: 0.153769 | Val MSE: 0.164243 | Train L0: 50.0 | Val L0: 50.0
 Época 158/1000 | Train MSE: 0.153778 | Val MSE: 0.164160 | Train L0: 50.0 | Val L0: 50.0
 Época 159/1000 | Train MSE: 0.153762 | Val MSE: 0.164113 | Train L0: 50.0 | Val L0: 50.0
 Época 160/1000 | Train MSE: 0.153757 | Val MSE: 0.164206 | Train L0: 50.0 | Val L0: 50.0
 Época 161/1000 | Train MSE: 0.153758 | Val MSE: 0.164131 | Train L0: 50.0 | Val L0: 50.0
 Época 162/1000 | Train MSE: 0.153738 | Val MSE: 0.164103 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 162 | Val MSE: 0.164103)
 Época 163/1000 | Train MSE: 0.153728 | Val MSE: 0.164196 | Train L0: 50.0 | Val L0: 50.0
 Época 164/1000 | Train MSE: 0.153727 | Val MSE: 0.164118 | Train L0: 50.0 | Val L0: 50.0
 Época 165/1000 | Train MSE: 0.153722 | Val MSE: 0.164137 | Train L0: 50.0 | Val L0: 50.0
 Época 166/1000 | Train MSE: 0.153715 | Val MSE: 0.164137 | Train L0: 50.0 | Val L0: 50.0
 Época 167/1000 | Train MSE: 0.153710 | Val MSE: 0.164126 | Train L0: 50.0 | Val L0: 50.0
 Época 168/1000 | Train MSE: 0.153702 | Val MSE: 0.164119 | Train L0: 50.0 | Val L0: 50.0
 Época 169/1000 | Train MSE: 0.153691 | Val MSE: 0.164210 | Train L0: 50.0 | Val L0: 50.0
 Época 170/1000 | Train MSE: 0.153687 | Val MSE: 0.164138 | Train L0: 50.0 | Val L0: 50.0
 Época 171/1000 | Train MSE: 0.153680 | Val MSE: 0.164161 | Train L0: 50.0 | Val L0: 50.0
 Época 172/1000 | Train MSE: 0.153671 | Val MSE: 0.164131 | Train L0: 50.0 | Val L0: 50.0
 Época 173/1000 | Train MSE: 0.153661 | Val MSE: 0.164196 | Train L0: 50.0 | Val L0: 50.0
 Época 174/1000 | Train MSE: 0.153659 | Val MSE: 0.164148 | Train L0: 50.0 | Val L0: 50.0
 Época 175/1000 | Train MSE: 0.153660 | Val MSE: 0.164245 | Train L0: 50.0 | Val L0: 50.0
 Época 176/1000 | Train MSE: 0.153647 | Val MSE: 0.164152 | Train L0: 50.0 | Val L0: 50.0
 Época 177/1000 | Train MSE: 0.153645 | Val MSE: 0.164261 | Train L0: 50.0 | Val L0: 50.0
 Época 178/1000 | Train MSE: 0.153631 | Val MSE: 0.164110 | Train L0: 50.0 | Val L0: 50.0
 Época 179/1000 | Train MSE: 0.153620 | Val MSE: 0.164119 | Train L0: 50.0 | Val L0: 50.0
 Época 180/1000 | Train MSE: 0.153616 | Val MSE: 0.164110 | Train L0: 50.0 | Val L0: 50.0
 Época 181/1000 | Train MSE: 0.153609 | Val MSE: 0.164127 | Train L0: 50.0 | Val L0: 50.0
 Época 182/1000 | Train MSE: 0.153605 | Val MSE: 0.164084 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 182 | Val MSE: 0.164084)

Época 183/1000 | Train MSE: 0.153598 | Val MSE: 0.164114 | Train L0: 50.0 | Val L0: 50.0
 Época 184/1000 | Train MSE: 0.153588 | Val MSE: 0.164077 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 184 | Val MSE: 0.164077)
 Época 185/1000 | Train MSE: 0.153575 | Val MSE: 0.164121 | Train L0: 50.0 | Val L0: 50.0
 Época 186/1000 | Train MSE: 0.153572 | Val MSE: 0.164079 | Train L0: 50.0 | Val L0: 50.0
 Época 187/1000 | Train MSE: 0.153564 | Val MSE: 0.164085 | Train L0: 50.0 | Val L0: 50.0
 Época 188/1000 | Train MSE: 0.153564 | Val MSE: 0.164079 | Train L0: 50.0 | Val L0: 50.0
 Época 189/1000 | Train MSE: 0.153564 | Val MSE: 0.164135 | Train L0: 50.0 | Val L0: 50.0
 Época 190/1000 | Train MSE: 0.153555 | Val MSE: 0.164084 | Train L0: 50.0 | Val L0: 50.0
 Época 191/1000 | Train MSE: 0.153552 | Val MSE: 0.164143 | Train L0: 50.0 | Val L0: 50.0
 Época 192/1000 | Train MSE: 0.153535 | Val MSE: 0.164122 | Train L0: 50.0 | Val L0: 50.0
 Época 193/1000 | Train MSE: 0.153520 | Val MSE: 0.164057 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 193 | Val MSE: 0.164057)
 Época 194/1000 | Train MSE: 0.153525 | Val MSE: 0.164065 | Train L0: 50.0 | Val L0: 50.0
 Época 195/1000 | Train MSE: 0.153514 | Val MSE: 0.164099 | Train L0: 50.0 | Val L0: 50.0
 Época 196/1000 | Train MSE: 0.153516 | Val MSE: 0.164056 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 196 | Val MSE: 0.164056)
 Época 197/1000 | Train MSE: 0.153520 | Val MSE: 0.164040 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 197 | Val MSE: 0.164040)
 Época 198/1000 | Train MSE: 0.153506 | Val MSE: 0.164061 | Train L0: 50.0 | Val L0: 50.0
 Época 199/1000 | Train MSE: 0.153497 | Val MSE: 0.164069 | Train L0: 50.0 | Val L0: 50.0
 Época 200/1000 | Train MSE: 0.153502 | Val MSE: 0.164075 | Train L0: 50.0 | Val L0: 50.0
 Época 201/1000 | Train MSE: 0.153498 | Val MSE: 0.164044 | Train L0: 50.0 | Val L0: 50.0
 Época 202/1000 | Train MSE: 0.153482 | Val MSE: 0.164059 | Train L0: 50.0 | Val L0: 50.0
 Época 203/1000 | Train MSE: 0.153483 | Val MSE: 0.164048 | Train L0: 50.0 | Val L0: 50.0
 Época 204/1000 | Train MSE: 0.153482 | Val MSE: 0.164052 | Train L0: 50.0 | Val L0: 50.0
 Época 205/1000 | Train MSE: 0.153473 | Val MSE: 0.164044 | Train L0: 50.0 | Val L0: 50.0
 Época 206/1000 | Train MSE: 0.153467 | Val MSE: 0.164179 | Train L0: 50.0 | Val L0: 50.0
 Época 207/1000 | Train MSE: 0.153472 | Val MSE: 0.164016 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 207 | Val MSE: 0.164016)
 Época 208/1000 | Train MSE: 0.153462 | Val MSE: 0.164047 | Train L0: 50.0 | Val L0: 50.0
 Época 209/1000 | Train MSE: 0.153453 | Val MSE: 0.164103 | Train L0: 50.0 | Val L0: 50.0
 Época 210/1000 | Train MSE: 0.153458 | Val MSE: 0.164043 | Train L0: 50.0 | Val L0: 50.0

Época 211/1000 | Train MSE: 0.153437 | Val MSE: 0.164087 | Train L0: 50.0 | Val L0: 50.0
 Época 212/1000 | Train MSE: 0.153444 | Val MSE: 0.164084 | Train L0: 50.0 | Val L0: 50.0
 Época 213/1000 | Train MSE: 0.153436 | Val MSE: 0.164128 | Train L0: 50.0 | Val L0: 50.0
 Época 214/1000 | Train MSE: 0.153431 | Val MSE: 0.164067 | Train L0: 50.0 | Val L0: 50.0
 Época 215/1000 | Train MSE: 0.153425 | Val MSE: 0.164011 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 215 | Val MSE: 0.164011)
 Época 216/1000 | Train MSE: 0.153429 | Val MSE: 0.164056 | Train L0: 50.0 | Val L0: 50.0
 Época 217/1000 | Train MSE: 0.153413 | Val MSE: 0.164017 | Train L0: 50.0 | Val L0: 50.0
 Época 218/1000 | Train MSE: 0.153416 | Val MSE: 0.164078 | Train L0: 50.0 | Val L0: 50.0
 Época 219/1000 | Train MSE: 0.153418 | Val MSE: 0.164023 | Train L0: 50.0 | Val L0: 50.0
 Época 220/1000 | Train MSE: 0.153415 | Val MSE: 0.164059 | Train L0: 50.0 | Val L0: 50.0
 Época 221/1000 | Train MSE: 0.153406 | Val MSE: 0.164035 | Train L0: 50.0 | Val L0: 50.0
 Época 222/1000 | Train MSE: 0.153397 | Val MSE: 0.164039 | Train L0: 50.0 | Val L0: 50.0
 Época 223/1000 | Train MSE: 0.153388 | Val MSE: 0.164174 | Train L0: 50.0 | Val L0: 50.0
 Época 224/1000 | Train MSE: 0.153392 | Val MSE: 0.164051 | Train L0: 50.0 | Val L0: 50.0
 Época 225/1000 | Train MSE: 0.153387 | Val MSE: 0.164024 | Train L0: 50.0 | Val L0: 50.0
 Época 226/1000 | Train MSE: 0.153380 | Val MSE: 0.164033 | Train L0: 50.0 | Val L0: 50.0
 Época 227/1000 | Train MSE: 0.153391 | Val MSE: 0.164004 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 227 | Val MSE: 0.164004)
 Época 228/1000 | Train MSE: 0.153369 | Val MSE: 0.164020 | Train L0: 50.0 | Val L0: 50.0
 Época 229/1000 | Train MSE: 0.153370 | Val MSE: 0.163994 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 229 | Val MSE: 0.163994)
 Época 230/1000 | Train MSE: 0.153365 | Val MSE: 0.164054 | Train L0: 50.0 | Val L0: 50.0
 Época 231/1000 | Train MSE: 0.153355 | Val MSE: 0.164007 | Train L0: 50.0 | Val L0: 50.0
 Época 232/1000 | Train MSE: 0.153359 | Val MSE: 0.164003 | Train L0: 50.0 | Val L0: 50.0
 Época 233/1000 | Train MSE: 0.153358 | Val MSE: 0.164003 | Train L0: 50.0 | Val L0: 50.0
 Época 234/1000 | Train MSE: 0.153350 | Val MSE: 0.164023 | Train L0: 50.0 | Val L0: 50.0
 Época 235/1000 | Train MSE: 0.153345 | Val MSE: 0.163984 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 235 | Val MSE: 0.163984)
 Época 236/1000 | Train MSE: 0.153340 | Val MSE: 0.163997 | Train L0: 50.0 | Val L0: 50.0
 Época 237/1000 | Train MSE: 0.153336 | Val MSE: 0.163987 | Train L0: 50.0 | Val L0: 50.0
 Época 238/1000 | Train MSE: 0.153335 | Val MSE: 0.163992 | Train L0: 50.0 | Val L0: 50.0
 Época 239/1000 | Train MSE: 0.153326 | Val MSE: 0.164018 | Train L0: 50.0 | Val L0: 50.0
 Época 240/1000 | Train MSE: 0.153329 | Val MSE: 0.163974 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 240 | Val MSE: 0.163974)

Época 241/1000		Train MSE: 0.153319		Val MSE: 0.164051		Train L0: 50.0		Val L0: 50.0
Época 242/1000		Train MSE: 0.153319		Val MSE: 0.164018		Train L0: 50.0		Val L0: 50.0
Época 243/1000		Train MSE: 0.153313		Val MSE: 0.164129		Train L0: 50.0		Val L0: 50.0
Época 244/1000		Train MSE: 0.153315		Val MSE: 0.164058		Train L0: 50.0		Val L0: 50.0
Época 245/1000		Train MSE: 0.153308		Val MSE: 0.164017		Train L0: 50.0		Val L0: 50.0
Época 246/1000		Train MSE: 0.153305		Val MSE: 0.164033		Train L0: 50.0		Val L0: 50.0
Época 247/1000		Train MSE: 0.153302		Val MSE: 0.163973		Train L0: 50.0		Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 247 | Val MSE: 0.163973)

Época 248/1000		Train MSE: 0.153292		Val MSE: 0.164004		Train L0: 50.0		Val L0: 50.0
Época 249/1000		Train MSE: 0.153301		Val MSE: 0.164015		Train L0: 50.0		Val L0: 50.0
Época 250/1000		Train MSE: 0.153284		Val MSE: 0.163982		Train L0: 50.0		Val L0: 50.0
Época 251/1000		Train MSE: 0.153290		Val MSE: 0.163950		Train L0: 50.0		Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 251 | Val MSE: 0.163950)

Época 252/1000		Train MSE: 0.153280		Val MSE: 0.164017		Train L0: 50.0		Val L0: 50.0
Época 253/1000		Train MSE: 0.153277		Val MSE: 0.163972		Train L0: 50.0		Val L0: 50.0
Época 254/1000		Train MSE: 0.153271		Val MSE: 0.164007		Train L0: 50.0		Val L0: 50.0
Época 255/1000		Train MSE: 0.153267		Val MSE: 0.163981		Train L0: 50.0		Val L0: 50.0
Época 256/1000		Train MSE: 0.153270		Val MSE: 0.164000		Train L0: 50.0		Val L0: 50.0
Época 257/1000		Train MSE: 0.153262		Val MSE: 0.163994		Train L0: 50.0		Val L0: 50.0
Época 258/1000		Train MSE: 0.153248		Val MSE: 0.164040		Train L0: 50.0		Val L0: 50.0
Época 259/1000		Train MSE: 0.153251		Val MSE: 0.163963		Train L0: 50.0		Val L0: 50.0
Época 260/1000		Train MSE: 0.153246		Val MSE: 0.163951		Train L0: 50.0		Val L0: 50.0
Época 261/1000		Train MSE: 0.153246		Val MSE: 0.164010		Train L0: 50.0		Val L0: 50.0
Época 262/1000		Train MSE: 0.153249		Val MSE: 0.163997		Train L0: 50.0		Val L0: 50.0
Época 263/1000		Train MSE: 0.153235		Val MSE: 0.164027		Train L0: 50.0		Val L0: 50.0
Época 264/1000		Train MSE: 0.153244		Val MSE: 0.164105		Train L0: 50.0		Val L0: 50.0
Época 265/1000		Train MSE: 0.153237		Val MSE: 0.164103		Train L0: 50.0		Val L0: 50.0
Época 266/1000		Train MSE: 0.153223		Val MSE: 0.164068		Train L0: 50.0		Val L0: 50.0
Época 267/1000		Train MSE: 0.153233		Val MSE: 0.164019		Train L0: 50.0		Val L0: 50.0
Época 268/1000		Train MSE: 0.153223		Val MSE: 0.163989		Train L0: 50.0		Val L0: 50.0
Época 269/1000		Train MSE: 0.153212		Val MSE: 0.163940		Train L0: 50.0		Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_to

topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt

Mejor modelo guardado (época 269 | Val MSE: 0.163940)

Época 270/1000		Train MSE: 0.153208		Val MSE: 0.163953		Train L0: 50.0		Val L0: 50.0
Época 271/1000		Train MSE: 0.153210		Val MSE: 0.163942		Train L0: 50.0		Val L0: 50.0
Época 272/1000		Train MSE: 0.153202		Val MSE: 0.163950		Train L0: 50.0		Val L0: 50.0

Época 273/1000 | Train MSE: 0.153204 | Val MSE: 0.164029 | Train L0: 50.0 | Val L0: 50.0
 Época 274/1000 | Train MSE: 0.153206 | Val MSE: 0.163957 | Train L0: 50.0 | Val L0: 50.0
 Época 275/1000 | Train MSE: 0.153200 | Val MSE: 0.163967 | Train L0: 50.0 | Val L0: 50.0
 Época 276/1000 | Train MSE: 0.153189 | Val MSE: 0.163935 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 276 | Val MSE: 0.163935)
 Época 277/1000 | Train MSE: 0.153185 | Val MSE: 0.164020 | Train L0: 50.0 | Val L0: 50.0
 Época 278/1000 | Train MSE: 0.153187 | Val MSE: 0.163946 | Train L0: 50.0 | Val L0: 50.0
 Época 279/1000 | Train MSE: 0.153189 | Val MSE: 0.163921 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 279 | Val MSE: 0.163921)
 Época 280/1000 | Train MSE: 0.153181 | Val MSE: 0.163933 | Train L0: 50.0 | Val L0: 50.0
 Época 281/1000 | Train MSE: 0.153173 | Val MSE: 0.163903 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 281 | Val MSE: 0.163903)
 Época 282/1000 | Train MSE: 0.153168 | Val MSE: 0.163957 | Train L0: 50.0 | Val L0: 50.0
 Época 283/1000 | Train MSE: 0.153159 | Val MSE: 0.163929 | Train L0: 50.0 | Val L0: 50.0
 Época 284/1000 | Train MSE: 0.153162 | Val MSE: 0.163911 | Train L0: 50.0 | Val L0: 50.0
 Época 285/1000 | Train MSE: 0.153164 | Val MSE: 0.163956 | Train L0: 50.0 | Val L0: 50.0
 Época 286/1000 | Train MSE: 0.153162 | Val MSE: 0.163991 | Train L0: 50.0 | Val L0: 50.0
 Época 287/1000 | Train MSE: 0.153154 | Val MSE: 0.163909 | Train L0: 50.0 | Val L0: 50.0
 Época 288/1000 | Train MSE: 0.153139 | Val MSE: 0.163921 | Train L0: 50.0 | Val L0: 50.0
 Época 289/1000 | Train MSE: 0.153137 | Val MSE: 0.163936 | Train L0: 50.0 | Val L0: 50.0
 Época 290/1000 | Train MSE: 0.153149 | Val MSE: 0.163918 | Train L0: 50.0 | Val L0: 50.0
 Época 291/1000 | Train MSE: 0.153140 | Val MSE: 0.163892 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 291 | Val MSE: 0.163892)
 Época 292/1000 | Train MSE: 0.153132 | Val MSE: 0.163948 | Train L0: 50.0 | Val L0: 50.0
 Época 293/1000 | Train MSE: 0.153126 | Val MSE: 0.163901 | Train L0: 50.0 | Val L0: 50.0
 Época 294/1000 | Train MSE: 0.153124 | Val MSE: 0.163909 | Train L0: 50.0 | Val L0: 50.0
 Época 295/1000 | Train MSE: 0.153130 | Val MSE: 0.163928 | Train L0: 50.0 | Val L0: 50.0
 Época 296/1000 | Train MSE: 0.153127 | Val MSE: 0.163967 | Train L0: 50.0 | Val L0: 50.0
 Época 297/1000 | Train MSE: 0.153120 | Val MSE: 0.163880 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\12500games\sae_topk_batch-topk_l6_12500g_ex1_best.pt
 Mejor modelo guardado (época 297 | Val MSE: 0.163880)
 Época 298/1000 | Train MSE: 0.153116 | Val MSE: 0.163912 | Train L0: 50.0 | Val L0: 50.0
 Época 299/1000 | Train MSE: 0.153105 | Val MSE: 0.163905 | Train L0: 50.0 | Val L0: 50.0
 Época 300/1000 | Train MSE: 0.153109 | Val MSE: 0.163934 | Train L0: 50.0 | Val L0: 50.0

Época 301/1000		Train MSE: 0.153112		Val MSE: 0.163897		Train L0: 50.0		Val L0: 50.0
Época 302/1000		Train MSE: 0.153108		Val MSE: 0.163904		Train L0: 50.0		Val L0: 50.0
Época 303/1000		Train MSE: 0.153101		Val MSE: 0.163912		Train L0: 50.0		Val L0: 50.0
Época 304/1000		Train MSE: 0.153104		Val MSE: 0.163964		Train L0: 50.0		Val L0: 50.0
Época 305/1000		Train MSE: 0.153100		Val MSE: 0.163940		Train L0: 50.0		Val L0: 50.0
Época 306/1000		Train MSE: 0.153102		Val MSE: 0.163903		Train L0: 50.0		Val L0: 50.0
Época 307/1000		Train MSE: 0.153097		Val MSE: 0.163967		Train L0: 50.0		Val L0: 50.0
Época 308/1000		Train MSE: 0.153091		Val MSE: 0.163932		Train L0: 50.0		Val L0: 50.0
Época 309/1000		Train MSE: 0.153082		Val MSE: 0.163896		Train L0: 50.0		Val L0: 50.0
Época 310/1000		Train MSE: 0.153082		Val MSE: 0.163955		Train L0: 50.0		Val L0: 50.0
Época 311/1000		Train MSE: 0.153093		Val MSE: 0.163888		Train L0: 50.0		Val L0: 50.0
Época 312/1000		Train MSE: 0.153086		Val MSE: 0.163923		Train L0: 50.0		Val L0: 50.0
Época 313/1000		Train MSE: 0.153080		Val MSE: 0.163885		Train L0: 50.0		Val L0: 50.0
Época 314/1000		Train MSE: 0.153076		Val MSE: 0.163906		Train L0: 50.0		Val L0: 50.0
Época 315/1000		Train MSE: 0.153064		Val MSE: 0.163891		Train L0: 50.0		Val L0: 50.0
Época 316/1000		Train MSE: 0.153062		Val MSE: 0.163909		Train L0: 50.0		Val L0: 50.0
Época 317/1000		Train MSE: 0.153066		Val MSE: 0.163906		Train L0: 50.0		Val L0: 50.0

Early stopping en época 317 (mejor: época 297)

Mejor época: 297 | Mejor Val MSE: 0.163880

Threshold de inferencia estimado: 0.0020

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
```

```

    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_16_12500g_ex1_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_16_12500games\sae_topk_batch-topk_16_12500g_ex1_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk_batch-topk_16_12500games/sae_topk_batch-topk_16_12500g_ex1_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],

```

```

NAMING_META['capa'],
NAMING_META['juegos'],
extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
    'val_l0_final': history['val_l0_sparsity'][-1],
    'val_fraction_active_final': history['val_fraction_active'][-1],
    # Contexto del entrenamiento
    'best_epoch': best_epoch,
    'total_epochs': len(history['train_mse_loss']),
    'train_mse_final': history['train_mse_loss'][-1],
    'train_l0_final': history['train_l0_sparsity'][-1],
    # Hiperparámetros TopK
    'architecture': cfg["architecture"],
    'k': cfg["k"],
    'd_sae': cfg["d_sae"],
    'd_in': cfg["d_in"],
    'expansion_factor': cfg["d_sae"] // cfg["d_in"],
    'hook_name': cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae\topk_topk\12500games\ex1\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.163906
val_mse_best	0.16388
val_l0_final	50.0
val_fraction_active_final	0.012207

```

        best_epoch                297
        total_epochs              317
        train_mse_final           0.153066
        train_l0_final            50.0
        architecture              batchtopk
        k                        50
        d_sae                     4096
        d_in                      512
        expansion_factor          8
        hook_name blocks.5.hook_resid_post

```

10. Visualización de Resultados

```

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]

```

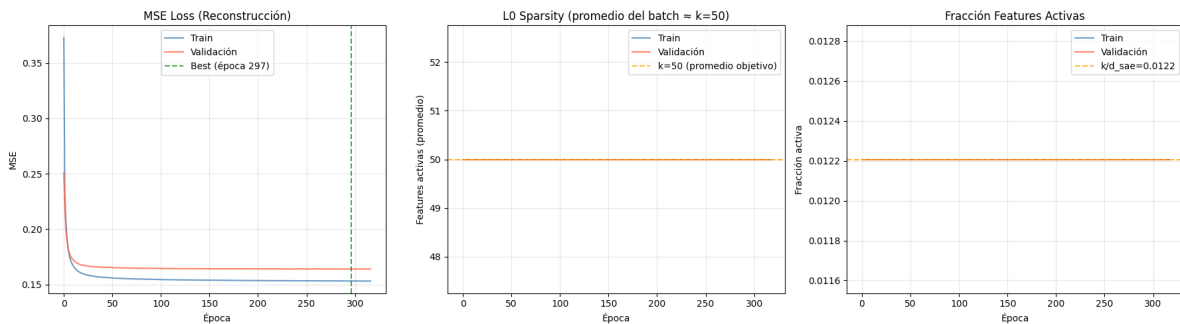
```

axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                 label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



=====
Gráficas guardadas exitosamente
=====

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```

import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],

```



```

        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\12500games\ex1\sae_topk_batch-topk_l6_12500g_ex1_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\12500games\ex1\sae_topk_batch-topk_l6_12500g_ex1_training.pdf